Supplementary Material for: Neuro-Symbolic Instruction Tuning for Explainable Mahjong Agents via Two-Stage Dual-LoRA

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A Neuro-Symbolic Knowledge Injection Details

In the main paper, we formulated the efficiency metrics. In this section, we provide the formal definitions and heuristic formulations for the safety and situation metrics used in our symbolic calculator $\mathcal{K}$.

A.1 Safety

While efficiency focuses on the ego-centric hand, safety metrics evaluate the risk of dealing into an opponent's hand using only observable information. We formalize defensive risk by analyzing the relationship between a candidate discard a and the observable constraints of each opponent p. Let $\mathcal{D}_p$ denote the set of tiles discarded by opponent p.

Genbutsu *Genbutsu* ($\mathcal{G}_p$) refers to the set of absolutely safe tiles against opponent p, derived from the *Furiten* rule in Riichi Mahjong. A tile a is strictly safe against player p if it exists in p's discard history:

$$a \in \mathcal{G}_p \iff a \in \mathcal{D}_p \quad (\implies P_{\text{deal-in}}(a, p) = 0). \tag{1}$$

This is the only safety metric that provides a binary and deterministic guarantee.

Suji *Suji* structures identify semi-safe tiles based on the topology of sequence waits. For simple two-sided waits (Ryanmen), a tile a is considered *Suji-safe* against opponent p if one of its structural relatives has appeared in $\mathcal{D}_p$:

$$\text{IsSuji}(a, p) = \mathbb{I}\left((a - 3 \in \mathcal{D}_p) \vee (a + 3 \in \mathcal{D}_p)\right). \tag{2}$$

While not providing absolute guarantees, *Suji*-safe tiles exhibit a lower empirical deal-in rate than random non-safe tiles.

Kabe Based on *Kabe* theory, a potential wait on tile a becomes logically infeasible if the adjacent tiles required to form a sequence wait are fully exhausted. We define:

$$\text{IsBlocked}(a) = \mathbb{I}\left(N_{\text{vis}}(k) = 4 \text{ for any } k \text{ essential for } a\right). \tag{3}$$

This metric captures safety arising from physical impossibility rather than opponent intent.

Live Tile (*Namigata*) A tile a is considered *Live* if it has never appeared in any opponent's discard pile, indicating maximal uncertainty about opponents' waiting patterns:

$$\text{Live}(a) \iff N_{\text{disc}}^{\text{opp}}(a) = 0, \tag{4}$$

where $N_{\text{disc}}^{\text{opp}}(a)$ denotes the number of times tile a appears in all opponents' discard histories.

Exhausted Tile (*Dead Tile*) A tile a is considered *Exhausted* if all four instances are already observable in the game state, making it physically impossible for opponents to be waiting on it:

$$\text{Exhausted}(a) \iff N_{\text{vis}}(a) \geq 4. \tag{5}$$

These exposure states are symbolic indicators derived from observable information and are used as heuristic cues for defensive risk estimation rather than absolute guarantees.

A.2 Situation

Optimal discard decisions are often governed by the perceived likelihood that opponents are already in *Tenpai* (ready-to-win). In practice, players dynamically adjust the relative emphasis between offensive efficiency and defensive safety based on this belief. To model this behavior, we introduce an accumulative heuristic scoring function that estimates the *Tenpai* probability of an opponent p, denoted as $P_{\text{tenpai}}(p)$.

The score is initialized based on visible commitments (such as open melds N_{meld}) and dynamically adjusted by game progression and special events:

$$P_{\text{tenpai}}(p) \propto \text{Base}(N_{\text{meld}}) + \delta_{\text{turn}} \cdot \mathbb{I}(N_{\text{turn}} > \tau) + \delta_{\text{risk}} \cdot \mathbb{I}_{\text{dora}}(d_{\text{recent}}), \tag{6}$$

where:

- $\text{Base}(N_{\text{meld}})$ assigns a baseline probability (e.g., 0 melds $\rightarrow$ Low, 3+ melds $\rightarrow$ High).
- δ_{turn} adds a penalty factor as the game progresses deep into the round (e.g., $N_{\text{turn}} > 12$).

– $\mathbb{I}_{\text{dora}}(d_{\text{recent}})$ triggers a risk increase if the opponent has recently discarded a *Dora* tile or a dangerous inner tile, often signaling that their hand is near completion.

The final estimated probability is discretized into four symbolic threat levels: `None`, `Low`, `Medium`, and `Riichi/High`, which are then injected into the LLM's prompt to modulate its aggressiveness.

B Case Study

In this section, we present three representative cases to illustrate the behavior of vanilla models, API-based models, and our fine-tuned models. To simplify the presentation, we select Qwen2.5-7B, DeepSeek-V3, and NeSy-Llama as representatives, as they achieve the strongest performance within their respective categories under the LLM-as-a-Judge evaluation.

Note that our experiments are conducted in Chinese. We provide full English translations here to ensure transparency.

B.1 Case 1: Early Game Efficiency (Calculation Check)

Input Game State

```
[Game Details]
Round: South 1, 0 Honba, 0 Riichi Sticks
Tiles Remaining: 60
Dealer (Oya): Player 0
Dora Indicator: 2p

[Your Status]
POV: Player 1 (South), Score: 34000
Hand: 2m 2m 3m 7m 7m 7p 8p 6s 7s Wh S G R 1p
Drawn Tile: 1p

[Table Status]
Player 0 (East, Dealer, 33400):
  Discards: N 9p 1m
Player 1 (South, You, 34000):
  Discards: N 9s
Player 2 (West, 9600):
  Discards: 1s E
Player 3 (North, 23000):
  Discards: 9p 9s
```

Model Outputs

Ground Truth Discard: 1p

Qwen2.5-7B: Discard: 7m

Reasoning: The reason for discarding 7m is: You already have three 7m tiles in your hand, so the probability of forming a sequence is extremely high. If other players need 7m to complete a meld or sequence, discarding this 7m will increase their difficulty, especially since the East player has already discarded 9p and might be looking for 7m to reach Tenpai. Additionally, discarding this tile prevents you from forming a concealed or exposed triplet, maintaining hand flexibility.

[Analysis]: Hallucination. The hand actually contains a pair of 7m (two tiles), not three. Breaking a pair here is strategically unsound.

DeepSeek-V3: Discard: 1p

Analysis: Discarding 1p is the optimal choice for the following reasons: 1. Safety Consideration: 1p is a Genbutsu (Safe Tile) (Opponent has discarded 1m), and 1s has also been discarded, so the safety factor is high. 2. Efficiency Analysis: Keeping 1p forms a "112p" bad shape, which has low efficiency. Discarding 1p leaves the hand with good shapes like 2277m/78p/67s, widening the subsequent tile acceptance (3-9m/6-9p/5-8s). 3. Defensive Advantage: As the South player with a 34,000 lead, maintaining a safe strategy is reasonable to avoid feeding the dangerous opponent (Player 2 with 9,600 points). Combining efficiency and defense, cutting 1p is the most balanced choice.

[Analysis]: Correct decision, but includes a factual error regarding safety rules (1p is not Genbutsu just because 1m was discarded).

NeSy-Llama (Ours): Discard: 1p

Reasoning: The hand is currently comparatively scattered, and the goal is to reach Tenpai quickly. Although 1p is a live tile, it is clearly the tile with the lowest efficiency in the hand. Other tiles have better connections, and keeping them allows for drawing more useful tiles. Since it is early game, the probability of others being in Tenpai is low, so safety is not the primary consideration. Discard 1p first and observe the situation to see if subsequent draws bring better changes.

[Analysis]: Correct. The model correctly identifies 1p as the inefficient isolated tile based on the symbolic calculation, without hallucinating safety properties.

B.2 Case 2: Late Game Defense (Safety Check)

Input Game State

```
[Game Details]
Round: East 2, 0 Honba, 1 Riichi Stick
Tiles Remaining: 19
Dealer (Oya): Player 1
Dora Indicator: 8p

[Your Status]
POV: Player 1 (East), Score: 25000
```

```
Hand: 4m 5m 5p 6p 9p 2s 4s 6s 7s S S N N 7m
Drawn Tile: 7m

[Table Status]
Player 0 (North, 22400, Riichi):
  Discards: W E G Wh R N 9s 8p 6m 5s 6m 1s* 1m
Player 1 (East, You, 25000):
  Discards: 1s 1m 8p 3p W 5m 9s 4m 3p 8m 1s 9s
Player 2 (South, 26600, Melds: Pon Wh):
  Discards: 1p 1p 1p 2s 4p E 3m 1m 6s G R 3m 2s
Player 3 (West, 25000):
  Discards: 1s 2m E G R 1m 9p Wh W 2p 2s 0m 5s
```

Model Outputs

Ground Truth Discard: 2s

Qwen2.5-7B: Discard: 5p

Reasoning: The reason for discarding 5p is: Your hand contains 5m, so discarding 5p can help form a sequence 5m-5p-5s. Additionally, 5p might be a tile needed by Player 2. At the same time, discarding 5p will not help opponents complete sequences or triplets, reducing the possibility of them reaching Tenpai. Therefore, discarding 5p is a relatively reasonable strategic choice.

[Analysis]: Severe Hallucination. "5m-5p-5s" is not a legal sequence in Mahjong.

DeepSeek-V3: Discard: 9p

Reasoning: Based on the current board analysis, discarding 9p is the optimal choice. Reasons: 1. The hand already has a pair of 7m, serving as the head. 2. 9p is an isolated tile with no adjacent shapes, having low retention value. 3. Opponents have discarded 8p (Dora indicator), so 9p has a relatively high safety factor. 4. Discarding 9p steers the hand toward high-efficiency sequence-based yaku (e.g., sanshoku doujun or ikkitsuukan), improving the likelihood of reaching Tenpai. 5. As the Dealer, one should prioritize winning speed. Combining efficiency and safety, cutting 9p is the most reasonable tactical choice.

[Analysis]: Risky. With 19 tiles left and an opponent in Riichi, discarding a live tile (9p) is dangerous. The model prioritizes offense too aggressively.

NeSy-Llama (Ours): Discard: 2s

Reasoning: It is currently late game, and the table situation is complex. The Upper player has declared Riichi, and the Lower player is clearly building a big hand; the probability of three players being in Tenpai is very high. Although discarding 7m, 9p, N, or S can maintain 3-shanten, 2s is a "Suji" tile against the Riichi player and is nearly exhausted, making it relatively safe. Although there is a slight loss in tile efficiency, considering the safety factors, prioritizing relatively safe tiles is the correct choice in the late game.

[Analysis]: Strategic Win. The model correctly identifies 2s as a

safe exit (Suji-trap against Riichi) derived from the symbolic "Situation" metrics.

B.3 Case 3: Human Alignment

Input Game State

```
[Game Details]
Round: South 4, 0 Honba, 0 Riichi Sticks
Tiles Remaining: 13
Dealer (Oya): Player 0
Dora Indicator: 3s

[Your Status]
POV: Player 0 (Dealer), Score: 32300
Hand: 4m 4m 0m 5m 6m 7m 7m 8m 9m 4s 6s E E 0s
Drawn Tile: 0s

[Table Status]
Player 0 (Dealer, You, 32300):
  Discards: 1s 9s S 2p 8s 7p 9m N 1p 1m 3p 5p 9p N
Player 1 (25500):
  Discards: W Wh S 1p N N S 9p E 7s 2p R 9s 1s
Player 2 (7900, Riichi):
  Discards: 1p 9p Wh G R 2p 2s 7s 2s 9p 6m G 3m 5p*
Player 3 (33300):
  Discards: W Wh 1p 2p G R 1s 2m 3m 3s 5p 3m 2s W
```

Model Outputs

Community Vote: 46% chose 5p, 38% chose 7m, 9% chose 5m (Push). Only a minority chose to fold (safe tiles).

Qwen2.5-7B: Discard: 4m

Reasoning: The reason for discarding 4m is: Your hand contains two 4m tiles, forming a pair, and currently there are no other players who need 4m to complete a sequence or triplet. Discarding 4m increases the difficulty for other players to form triplets, especially for the player in the winning seat, limiting their winning possibilities. At the same time, keeping 4m in hand is beneficial for your subsequent draw strategy.

[Analysis]: Weak reasoning. Breaking a pair in a ready hand is inefficient, and 4m is dangerous.

DeepSeek-V3: Discard: E (East)

Reasoning: According to the board analysis, discarding E is the optimal choice: 1. The hand already has a 0s pair, so the E pair is less useful, and E is a honor tile that is hard to form a sequence. 2. After discarding E, keeping 0s maintains a two-pair structure, offering a chance for Seven Pairs. 3. The discard river shows E has not been discarded in large numbers, but it has a relatively high safety factor. 4. Keeping continuous numbers like 0m-9m

makes it easier to form Mixed Triple Sequence or Pure Straight. 5. Opponent 2 has declared Riichi; discarding E is safer than discarding middle tiles. Combining efficiency and defense, cutting E is the most balanced choice.
[Analysis]: Good decision. DeepSeek correctly identifies the defensive value of East (E) against the Riichi.

NeSy-Llama (Ours): Discard: E (East)

Reasoning: It is currently late game, and the probability of everyone being in Tenpai is high, especially since the Lower player has declared Riichi. Although discarding 0m, 4m, 5m, or 7m allows us to reach Tenpai, they are all live tiles and carry extremely high risk. The most important thing now is safety. East (E) is a Genbutsu against the Upper player and is relatively safe. Although discarding East retreats from 1-shanten, preserving the score is more important. Dealing in now would end the game immediately. There is no need to risk the current rank for a Tenpai.
[Analysis]: Expert Judgment. NeSy-Llama explicitly accounts for the global game context (All Last, leading position) and chooses to fully defend rather than pursue Tenpai. This behavior aligns with professional game-theoretic decision-making, prioritizing score preservation over local tile efficiency.

C Detailed Data Preprocessing Steps

We utilized gameplay logs from the Tenhou platform [4], utilizing data acquisition scripts adapted from community resources [5]. To strictly prevent data leakage, we partitioned the dataset based on match timestamps in the year 2024. Specifically, logs from December 20 to December 29 were allocated for Stage 1 training; December 30 for Stage 1 validation; and December 31 for testing across both stages. Additionally, data from January 1, 2024, was reserved for the Stage 2 training set.

We applied rigorous filters to ensure high strategic quality. We exclusively selected logs from the "Four-Player South with Red Dora" mode where all four participants held a rank of 7-*dan* or higher.

For each match, we designated the ego-centric agent from the top-2 finishing players to capture winning behaviors. For any specific round, the target player is sampled probabilistically proportional to their final match scores. However, to filter out explicit errors, if a candidate player dealt into a winning hand (*Ron*) in the current round, we deterministically selected the other candidate as the ego-centric agent.

We decomposed each round into state-action pairs. Since players cannot discard volitionally after declaring *Riichi* (as the rules enforce automatic discards of non-winning tiles), we excluded all post-*Riichi* turns from the dataset.

[4] `https://tenhou.net/sc/raw/`
[5] `https://tinyurl.com/tenhou-data-process`

Stratified Sampling and Balancing. To mitigate the long-tail distribution inherent in gameplay data, we employed a stratified sampling strategy based on the Cartesian product of two dimensions:

1. **Strategic Context:** Divided into *Offense* (no opponent has declared Riichi) and *Defense* (at least one opponent has declared Riichi).
2. **Game Phase:** Categorized by the number of discards made by the ego-player: *Early* (0–5), *Mid* (6–11), and *Late* (≥ 12).

This results in $2 \times 3 = 6$ distinct buckets. The raw distribution exhibited significant imbalance. For instance, in the Stage 1 training set, the sample counts were:

- Early-Defense: 7,339
- Early-Offense: 247,614
- Mid-Defense: 46,259
- Mid-Offense: 126,611
- Late-Defense: 33,526
- Late-Offense: 28,327

Adhering to the short-board principle, we performed random down-sampling to align all buckets with the minority class (Early-Defense). Consequently, we obtained $7,339 \times 6 = 44,034$ samples for the Stage 1 training set. Following the same protocol, we curated 4,416 samples for validation, 4,896 for testing, and 5,544 for Stage 2 training.

D Implementation Details

In our experiments, we adopted distinct hyper-parameter strategies for the two stages to accommodate their specific objectives. Stage 1 focuses on rapid policy alignment using QLoRA to minimize memory usage, while Stage 2 employs a higher-rank LoRA on a bfloat16 base model to capture complex reasoning patterns. The detailed configurations are listed in Table 1.

Notably, in Stage 2, we froze the parameters of the base model and the Stage 1 adapter (decision_adapter) while only updating the parameters of the new Stage 2 adapter (explanation_adapter). This setup ensures that the reasoning training does not interfere with the already aligned decision boundaries.

During the inference phase, we employed dynamic adapter switching. The generation parameters were tuned differently for the decision and reasoning phases to balance precision and creativity.

For the discard decision in Stage 1, we prioritize deterministic and accurate outputs. Therefore, we utilized beam search to explore the most probable action paths.

- **Decoding Strategy:** Beam Search
- **Num Beams:** 5
- **Num Return Sequences:** 3 (To obtain Top-3 candidates)

Table 1. Hyper-parameter settings for Stage 1 (Decision) and Stage 2 (Reasoning).

Hyper-parameter	**Stage 1: Decision**	**Stage 2: Reasoning**
Base Model Precision	4-bit (NF4)	bfloat16
LoRA Method	QLoRA	Standard LoRA
LoRA Rank (r)	16	32
LoRA Alpha (α)	32	64
LoRA Dropout	0.05	0.05
Target Modules	All Linear Layers†	
Batch Size	32	16
Micro Batch Size	2	2
Gradient Accumulation	16	8
Learning Rate	3×10^{-4}	1×10^{-4}
LR Scheduler	Cosine	Cosine
Num Epochs	3	3
Max Sequence Length	1024	1024
Optimizer	AdamW	AdamW

†Includes: q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj.

– **Max New Tokens:** 64

For explanation generation in Stage 2, we aim for diverse and natural linguistic expression while adhering to the injected symbolic facts. We employed nucleus sampling (Top-p).

– **Decoding Strategy:** Sampling
– **Temperature:** 0.3 (Low temperature to reduce hallucinations)
– **Top-p:** 0.9
– **Max New Tokens:** 512

E Prompt Templates

In this section, we present the prompt templates used for both the proposed NeSy-Mahjong and the baseline models. Since the explanation generation task targets Chinese readers (consistent with the Tenhou dataset language context), the original prompts for explanation generation were constructed in Chinese. We present their English translations here for clarity.

E.1 Input Context Representation

To enable LLMs to process the structured game information, we serialize the game state s and the symbolic metrics k into a text-based format. The placeholders (e.g., `{Player Score}`) are replaced by actual values during inference.

Game State Segment (s) This segment is used as input for both Stage 1 (Decision) and Stage 2 (Reasoning).

```
[Game Details]:
Game:
{Round Name},
{Honba} Honba,
{Riichi Sticks} Riichi Sticks,
{Tiles Left} Left
Oya (Dealer): Player {Dealer Index}
Dora Indicators: {Dora Indicators}

--- Your Status ---
POV: Player {ID} (Score: {Score}) ({Wind Position})
Hand: {Hand Tiles}
Drawn Tile: {Draw}

--- Table Status ---
Player 0 (Score: {Score}) (East):
  Discards: {Discard History}
... (Repeated for Player 1, 2) ...
Player 3 (Score: {Score}) (North) (You):
  Discards: {Discard History}
```

Neuro-Symbolic Context Segment (k) This segment contains the objective metrics calculated by the symbolic calculator $\mathcal{K}$. It is appended to the input only during Stage 2 (Reasoning). Note that this information is only provided to the NeSy-Mahjong models and the teacher model during Stage 2, and is excluded from the prompts of baseline models.

```
[Objective Data & Comparison]
Global Context: {Tenpai Probabilities}
[Discard Options Table] (Sorted by efficiency):
- [Discard {Tile A}]: {Shanten} Shanten, {Ukeire} Ukeire | {Safety Info}
- [Discard {Tile B}]: {Shanten} Shanten, {Ukeire} Ukeire | {Safety Info}
...
```

E.2 Prompts for Decision (Stage 1)

For the decision task, the model is required to output the optimal discard tile.

NeSy-Mahjong (Fine-tuned) Since our models are instruction-tuned specifically for this format, we use a concise system instruction without formatting constraints.

> **System Instruction:** You are an expert Japanese Mahjong AI. Analyze the following game state and determine which tile to discard.

Baselines (Vanilla & Commercial API) For models without task-specific tuning, we add explicit formatting constraints to ensure the output can be parsed programmatically.

> **System Instruction:** You are an expert Japanese Mahjong AI. Analyze the following game state and determine which tile to discard. **No more than 150 words.** At the end of your analysis, please output your final decision on a new line starting with the symbol '$', like this: $ 4s

E.3 Prompts for Explanation (Stage 2)

For the reasoning task, the input consists of the *Game State Segment*, the *Neuro-Symbolic Context Segment*, and the *Target Decision*.

NeSy-Mahjong (Fine-tuned)

> **System Instruction:** You are a rigorous Mahjong tactical analyst. Please analyze why this tile should be discarded based on the game state and objective calculation data (including the discard comparison table).

Baselines (Vanilla & Commercial API) Similar to the decision phase, we impose length constraints on baseline models to prevent verbose generation.

> **System Instruction:** You are a rigorous Mahjong tactical analyst. Please analyze why this tile should be discarded based on the game state and objective calculation data (including the discard comparison table). Requirement: Do not exceed 150 words.